Temporal Correlations Reshape Causal Resources in Qubit Hamiltonian Recognition: Exact Two-Query General-Strategy Phase Diagram

Youfu Wang

2026-09-10

Affiliation: Independent Researcher, OCN Research
ORCID: 0009-0005-5998-6024
Contact author: youfuwang888@gmail.com

Archival version note (v1.1). This Zenodo release conservatively restates the weak-correlation result as a set of explicit constructive lower bounds. The current proof package establishes a zero onset threshold and positive feasible improvements of orders η^2, $\eta^{4/3}$, and η in the three angular regimes, but it does not contain matching upper bounds proving exact asymptotic equivalence. The exact full-SEQ reduction, exact GEN=SEQ closure, exact interior equality curve, and exact high-angle reentrant phase are unchanged. This version is a public preprint and has not undergone independent peer review.

Abstract

Hamiltonian recognition asks which candidate Hamiltonian generated an unknown quantum evolution when the evolution angle is not observed. We solve a binary two-query qubit setting in which the two unknown angles range from independently resampled to fully shared. A task-specific symmetrization theorem reduces the full sequential optimization exactly to a three-dimensional correlation elliptope. We then construct a nine-dimensional legal subset of the most general tester dual whose reduced semidefinite-program dual is exactly the same elliptope problem, proving that general strategies cannot outperform sequential ones throughout the full parameter domain. Independent query angles give no advantage over parallel processing, whereas arbitrarily weak temporal correlation can activate a globally optimal sequential advantage. We give explicit positive weak-correlation lower bounds of orders η^2 below 45°, $\eta^{4/3}$ at 45°, and η above 45°. The complete phase diagram contains a unique interior equality curve and a high-angle reentrant regime in which the advantage vanishes at both zero and full temporal correlation but is positive at every intermediate strength.

I. Introduction

Hamiltonian recognition is a discrete quantum-hypothesis-testing problem in which a black-box evolution is generated by one of several known Hamiltonians while the evolution angle is unknown. Zhu *et al.* recently formalized the task using quantum testers, proved exact optimality results for orthogonal qubit Hamiltonians, and treated general nonorthogonal directions numerically [1]. The tester hierarchy distinguishes parallel (PAR), sequential/adaptive (SEQ), and general (GEN) strategies, with GEN allowing the most general channel-to-probability transformations considered in that framework.

Adaptive advantage in channel discrimination is well established [2-4], and temporal memory can change which discrimination architecture is optimal [5,6]. Recent work has also highlighted multi-time quantum-comb structures for sensing temporal correlations directly [7]. These results motivate a sharper question: for a fixed nonorthogonal Hamiltonian-recognition task, how does the strength of cross-query temporal correlation change the exact operational value of causal processing?

We solve this question for two queries, equal priors, and an equal-gap binary qubit pair. After removing physically irrelevant identity shifts and making a common basis choice, the candidates are

$$H_{\pm} = \pm \sin(\beta/2)X + \cos(\beta/2)Z, \qquad 0 \le \beta \le \pi/2.$$

The two-query process interpolates between independently resampled unknown angles and a fully shared unknown angle,

$$\Omega_{\pm}(\eta) = (1-\eta)\Omega_{\pm}^{\mathrm{iid}} + \eta\Omega_{\pm}^{\mathrm{shared}}, \qquad 0 \le \eta \le 1.$$

We compare the hierarchy

$$\mathrm{PAR} \subset \mathrm{SEQ} \subset \mathrm{GEN}.$$

The first structural result is a global full-SEQ symmetrization theorem: the complete sequential optimization reduces without loss to a three-parameter deterministic comb whose positivity domain is exactly a three-dimensional correlation elliptope. The second structural result closes the remaining strategy hierarchy: an explicit nine-dimensional subset of the legal GEN dual is sufficient to reproduce the same elliptope optimization exactly. Consequently,

$$P_{\mathrm{GEN}}(\beta,\eta) = P_{\mathrm{SEQ}}(\beta,\eta)$$

over the full two-parameter domain.

The resulting PAR-versus-GEN phase diagram has three distinctive features. Arbitrarily weak temporal correlation can activate a globally optimal causal-processing advantage. The weak-correlation onset changes nonanalytically at $\beta = 45^{\circ}$, where the leading scaling is $\eta^{4/3}$. Finally, the phase diagram contains both a unique interior equality curve and a high-angle reentrant phase in which the advantage vanishes at zero and full temporal correlation but is strictly positive for every intermediate strength.

II. Model and tester classes

For a normalized Pauli Hamiltonian $H^2 = I$,

$$U_H(\theta) = e^{-i\theta H} = \cos\theta\, I - i\sin\theta\, H,$$

with θ uniformly distributed on $[0,\pi]$. The two hypotheses have equal prior probability.

The temporal-correlation interpolation has the operational meaning:

- $\eta = 0$: the unknown angle is independently resampled for the two queries;
- $\eta = 1$: both queries share the same unknown angle;
- $0 < \eta < 1$: a convex mixture of these two processes.

For a binary two-slot sequential tester,

$$T_+ \succeq 0, \qquad T_- \succeq 0, \qquad W = T_+ + T_-.$$

Using Hilbert-space order (I_1, O_1, I_2, O_2), deterministic SEQ normalization is

$$W = I_{O_2} \otimes \Xi,$$

$$\mathrm{Tr}_{I_2}\,\Xi = \rho_{I_1} \otimes I_{O_1}, \qquad \rho \succeq 0, \quad \mathrm{Tr}\,\rho = 1.$$

The GEN tester class is defined by the corresponding general normalization over products of valid channel Choi operators. Its dual affine space is the Choi affine space of two-partite non-signalling channels without imposing positivity on the affine representative, as reviewed in Ref. [1] and the Supplemental Material.

III. Full sequential optimization reduces to a correlation elliptope

Theorem 1 (global full-SEQ reduction).
For every

$$0 \le \beta \le \pi/2, \qquad 0 \le \eta \le 1,$$

an optimal full sequential deterministic tester can be chosen in the three-parameter family

$$W_3 = \frac{1}{4} I^{\otimes 4} + a(XIXI + IXXI) + b(YIYI + IYYI) + d(ZIZI + IZZI).$$

Defining

$$x = 8a, \qquad y = 8b, \qquad z = 8d,$$

positivity is exactly equivalent to

$$G(x, y, z) = \begin{pmatrix} 1 & x & z \\ x & 1 & -y \\ z & -y & 1 \end{pmatrix} \succeq 0.$$

Therefore

$$P_{\mathrm{SEQ}}(\beta, \eta) = \max_{G(x,y,z)\succeq 0} P_\eta(x, y, z)$$

with

$$P_\eta(x, y, z) = \frac{1}{2} + A_\eta(x, z) + B_\eta(x, y, z),$$

where

$$A_\eta(x, z) = \frac{\sin\beta}{8} \left[\sqrt{(1+x)(1+z)} + (1-\eta)\sqrt{(1-x)(1-z)} \right],$$

$$B_\eta(x, y, z) = \frac{\sin\beta}{16} \sqrt{2(1+y)\, v_\eta^T G v_\eta}.$$

where

$$v_\eta = \begin{pmatrix} -\eta \\ \frac{\eta+2}{2}(1-\cos\beta) \\ \frac{\eta+2}{2}(1+\cos\beta) \end{pmatrix}.$$

The proof proceeds in two stages. First, exact task symmetries and the deterministic causal normalization reduce the full SEQ comb to six Pauli coefficients,

$$IIII,\ XIXI,\ IXXI,\ YIYI,\ IYYI,\ ZIZI,\ IZZI.$$

Second, after this legal reduction only, the task-specific $I_1 \leftrightarrow O_1$ exchange maps the six-parameter causal family into itself and leaves the performance operators invariant. Convex averaging therefore enforces pairwise coefficient equality. The characteristic polynomial of W_3 reduces positivity exactly to $G \succeq 0$. Full details are given in the Supplemental Material.

The parallel feasible set is the Bell tetrahedron embedded inside this elliptope. The PAR-SEQ resource question therefore becomes geometric: when does the correlation-dependent objective detect elliptope directions unavailable to the tetrahedron?

IV. General strategies do not outperform sequential strategies

The mother-paper GEN dual for equal priors can be written, after absorbing the scalar dual variable into Y, as

$$P_{\mathrm{GEN}} = \min_{Y} \frac{\mathrm{Tr}\, Y}{4}$$

subject to

$$2Y - \Omega_{+} \succeq 0, \qquad 2Y - \Omega_{-} \succeq 0,$$

and the two-slot non-signalling affine conditions

$$_{O_1}Y = {}_{I_1O_1}Y, \qquad {}_{O_2}Y = {}_{I_2O_2}Y.$$

Theorem 2 (global GEN-SEQ equality).
For the complete two-query family,

$$P_{\mathrm{GEN}}(\beta, \eta) = P_{\mathrm{SEQ}}(\beta, \eta)$$

for all

$$0 \le \beta \le \pi/2, \qquad 0 \le \eta \le 1.$$

Proof sketch. We explicitly construct a nine-dimensional legal subset of the GEN dual affine space. In a normalized Pauli-vector sector basis, every certificate in this subset is parametrized by one real symmetric matrix

$$K = \begin{pmatrix} a & u & v \\ u & b & w \\ v & w & c \end{pmatrix}$$

together with three scalar pairs related by

$$q_{-} - q_{+} = 2v, \qquad p_{-} - p_{+} = 2u, \qquad r_{-} - r_{+} = -2w.$$

The resulting restricted GEN certificate SDP is strictly feasible. Dualizing that finite-dimensional SDP gives a 3×3 PSD Lagrange block

$$G(x, y, z) = \begin{pmatrix} 1 & x & z \\ x & 1 & -y \\ z & -y & 1 \end{pmatrix} \succeq 0.$$

Optimizing the remaining dual off-diagonal entries produces exactly the objective $P_{\eta}(x, y, z)$ of Theorem 1. Hence

$$\min(\text{restricted GEN dual}) = \max_{G \succeq 0} P_{\eta} = P_{\mathrm{SEQ}}.$$

Because the restricted certificate family is a subset of valid full-GEN dual certificates, it provides the upper bound

$$P_{\mathrm{GEN}} \leq P_{\mathrm{SEQ}}.$$

Conversely SEQ is a valid subclass of GEN, so

$$P_{\mathrm{GEN}} \geq P_{\mathrm{SEQ}}.$$

The two inequalities prove equality. The nine explicit GEN generators, sector decomposition, strong-duality argument, and coefficient matching are given in the Supplemental Material.

Theorem 2 is important operationally: every strict sequential advantage derived below is already optimal against the most general tester class in this framework, including strategies beyond fixed causal order.

V. Independent-angle control

Define the globally optimal causal-processing gap relative to PAR,

$$\Delta(\beta,\eta) = P_{\mathrm{GEN}}(\beta,\eta) - P_{\mathrm{PAR}}(\beta,\eta) = P_{\mathrm{SEQ}}(\beta,\eta) - P_{\mathrm{PAR}}(\beta,\eta).$$

Theorem 3 (iid equality).
For independently resampled query angles,

$$P_{\mathrm{GEN}}(\beta,0) = P_{\mathrm{SEQ}}(\beta,0) = P_{\mathrm{PAR}}(\beta,0)$$

for all β, with exact success probability

For the lower-angle branch,

$$P^{(2)}_{\mathrm{iid}}(\beta) = \frac{1}{2} + \frac{\sin\beta(1+\cos\beta)}{4}, \qquad 0 \leq \beta \leq \pi/4.$$

For the upper-angle branch,

$$P^{(2)}_{\mathrm{iid}}(\beta) = \frac{5}{8} + \frac{\sin\beta}{4}, \qquad \pi/4 \leq \beta \leq \pi/2.$$

The transition occurs at 45°. This exact boundary shows that noncommutativity of the local Hamiltonians alone is insufficient to create a general-strategy advantage over PAR.

VI. Weak-correlation activation

Theorem 4 (zero onset threshold and constructive lower bounds).
For every

$$0 < \beta < \pi/2,$$

the onset threshold is

$$\eta_{\mathrm{onset}}(\beta) = 0.$$

More explicitly, the proof constructs legal SEQ points that imply the following positive weak-correlation lower bounds.

Below 45°,

$$\Delta(\beta,\eta) \geq K_<(\beta)\eta^2 + O(\eta^3), \qquad 0 < \beta < \pi/4,$$

with $K_<(\beta) > 0$.

At 45°,

$$\Delta(\pi/4,\eta) \geq K_c\eta^{4/3} + o(\eta^{4/3}),$$

with $K_c > 0$.

Above 45°,

$$\Delta(\beta,\eta) \geq K_>(\beta)\eta + O(\eta^2), \qquad \pi/4 < \beta < \pi/2,$$

with $K_>(\beta) > 0$.

The explicit coefficients are given in the Supplemental Material. These bounds are sufficient to prove $\eta_{\text{onset}}(\beta) = 0$ for every nontrivial angle. They also show a change in the order of the proven constructive improvement at 45°. No matching upper bound is claimed here, and this archival version therefore does not assert the exact asymptotic equivalences $\Delta \sim K\eta^\alpha$.

VII. Exact interior equality curve

Let

$$\beta_c = \arccos(1/\sqrt{3}).$$

Theorem 5 (unique interior equality curve).
For

$$0 < \beta < \beta_c, \qquad 0 < \eta < 1,$$

the unique interior PAR-GEN equality curve is

$$\sin^2\beta = \frac{4\eta^2}{\eta^3 + \eta^2 + 4}.$$

On this curve,

$$P_{\text{GEN}} = P_{\text{SEQ}} = P_{\text{PAR}} = \frac{1}{2} + \frac{2\eta}{\eta^3 + \eta^2 + 4}.$$

The curve connects $(0,0)$ continuously to $(\beta_c, 1)$.

Figure 1. Exact PAR-GEN phase-diagram boundaries. The solid curve is the unique interior equality curve. The dashed vertical line at 45° marks the change in the exponent of the proven constructive weak-correlation lower bound; it is not asserted here as an exact asymptotic critical exponent. The dotted vertical line marks $\beta_c = \arccos(1/\sqrt{3})$.

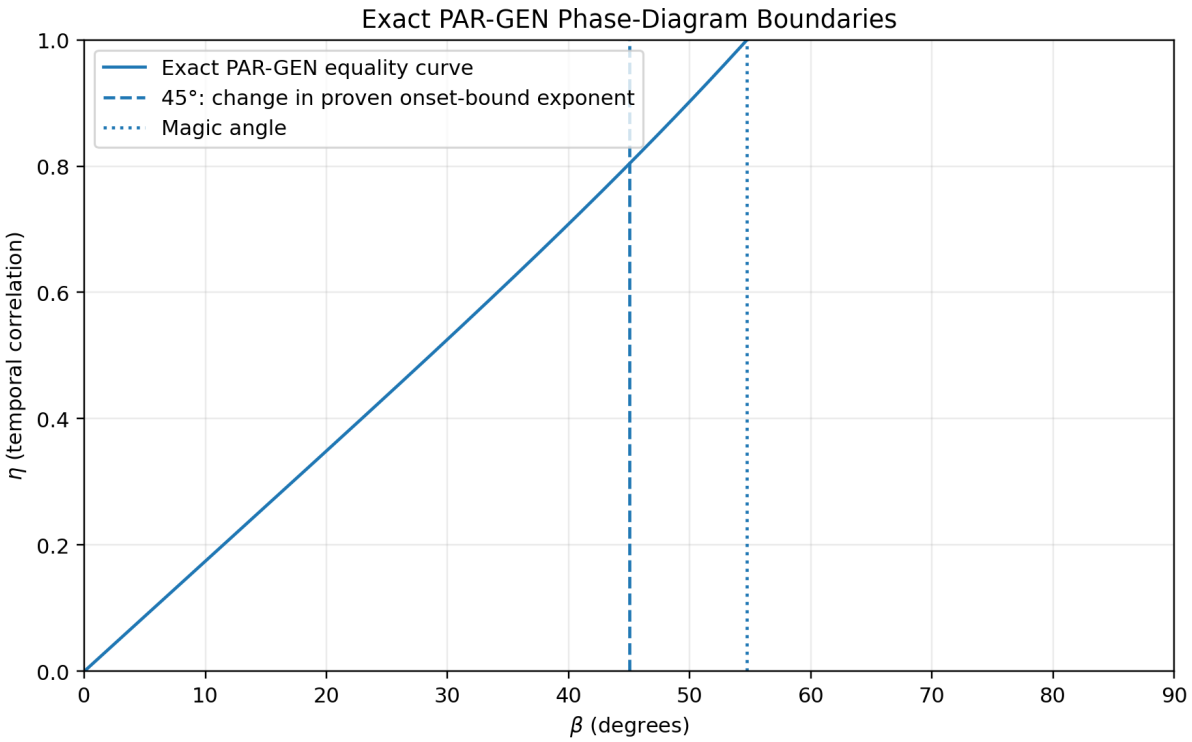


Figure 1: Exact PAR-GEN phase-diagram boundaries.

VIII. High-angle reentrant phase

Theorem 6 (reentrant temporal-correlation resource phase).
For

$$\beta_c \leq \beta < \pi/2,$$

the exact general-strategy gap satisfies

$$\Delta(\beta, 0) = 0,$$

$$\Delta(\beta, \eta) > 0 \qquad (0 < \eta < 1),$$

and

$$\Delta(\beta, 1) = 0.$$

Thus the globally optimal advantage follows

$$0 \to + \to 0.$$

The operational advantage is therefore nonmonotonic as a function of the temporal-correlation parameter in this task: correlation can activate causal-processing value and later remove it again.

At $\beta = 60^\circ$, independent numerical optimization over the Bell tetrahedron and the full sequential elliptope provides a direct cross-check. Theorem 2 identifies the latter with the GEN optimum, so the same numerical slice is a cross-check of the full PAR-GEN phase topology.

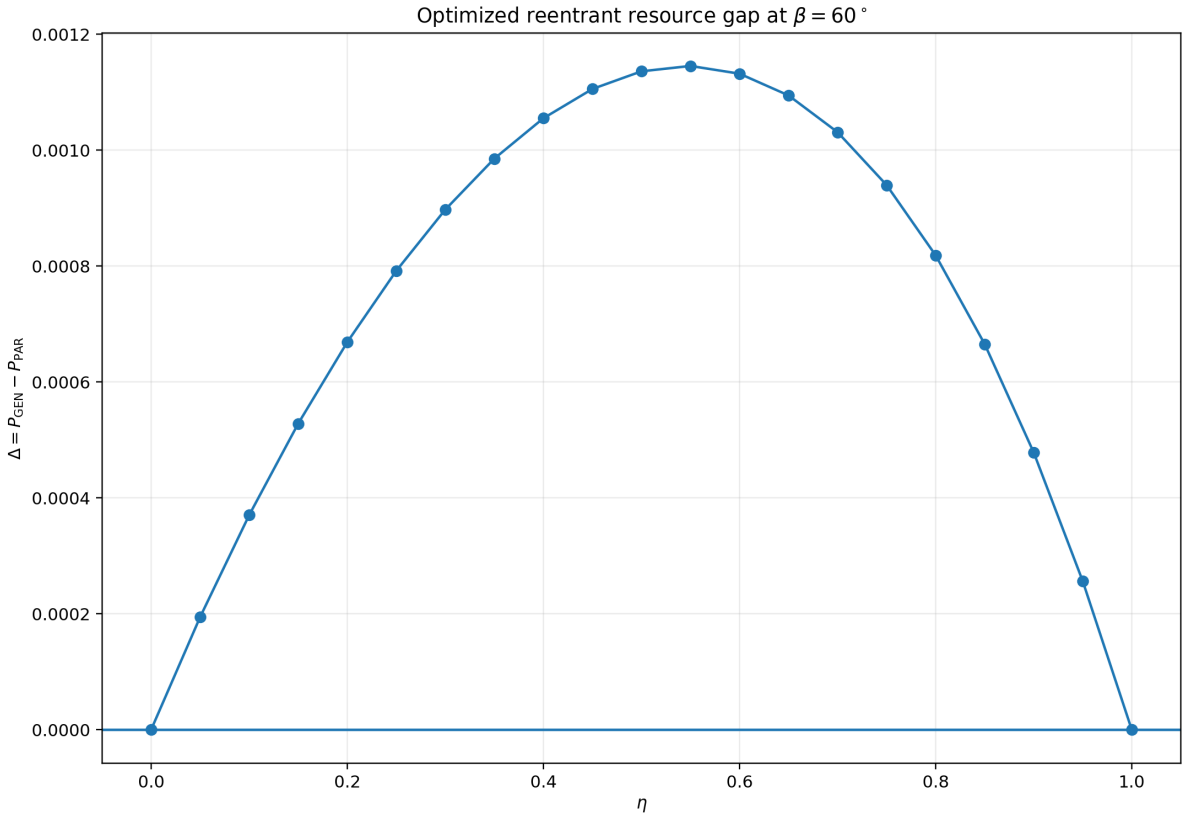


Figure 2: Numerically optimized reentrant GEN-PAR resource gap at beta=60 degrees.

Figure 2. Numerical cross-check of the reentrant GEN-PAR gap at $\beta = 60^\circ$. The interior gap is positive while the endpoint values vanish to numerical precision. Numerical optimization is not used to establish the analytic theorem.

IX. Shared-angle and orthogonal boundaries

Theorem 7 (fully shared and orthogonal boundaries).
At $\eta = 1$,

$$0 < \beta < \beta_c \quad \Longrightarrow \quad P_{\mathrm{GEN}} = P_{\mathrm{SEQ}} > P_{\mathrm{PAR}}.$$

At and above the magic angle,

$$\beta_c \le \beta \le \pi/2,$$

$$P_{\mathrm{GEN}} = P_{\mathrm{SEQ}} = P_{\mathrm{PAR}} = 5/6.$$

At the orthogonal endpoint,

$$\beta = \pi/2,$$

all three classes are equal for every η,

$$P_{\mathrm{GEN}} = P_{\mathrm{SEQ}} = P_{\mathrm{PAR}} = \frac{7 + 3\eta}{4(\eta + 2)}.$$

X. Complete two-query general-strategy phase diagram

Combining Theorems 2-7 gives the complete two-parameter PAR-versus-GEN resource phase diagram.

For the open interior

$$0 < \beta < \pi/2, \qquad 0 < \eta < 1,$$

one has

$$P_{\mathrm{GEN}} = P_{\mathrm{SEQ}} > P_{\mathrm{PAR}}$$

everywhere except on the unique low-angle equality curve of Theorem 5.

The boundary equalities are:

- $\eta = 0$: PAR = SEQ = GEN for all β;
- $\beta = 0$: PAR = SEQ = GEN for all η;
- $\beta = \pi/2$: PAR = SEQ = GEN for all η;
- $\eta = 1$: PAR = SEQ = GEN for $\beta \ge \beta_c$, with strict optimal SEQ/GEN separation below β_c.

XI. Interpretation

The Hamiltonian pair and its local noncommutativity are held fixed while η varies. Only the cross-query temporal structure changes. The results therefore rule out the simplified mechanism

$$\text{noncommutativity} \Rightarrow \text{adaptive advantage}.$$

Instead,

$$\text{temporal correlation changes which causal directions carry discrimination value}.$$

Geometrically, PAR is restricted to a Bell tetrahedron while the optimal sequential strategy explores a correlation elliptope. The GEN dual theorem shows that, for this task, going beyond sequential causal processing does not enlarge the attainable optimum: the same elliptope already captures the full general-strategy value.

This produces a concrete separation between two questions that are often conflated: whether a larger formal strategy class exists, and whether that larger class carries additional operational value for the task. Here GEN is formally larger than SEQ, but its extra degrees of freedom have zero discrimination value.

XII. Discussion

The iid boundary proves that causal adaptivity alone does not guarantee an advantage. Infinitesimal temporal correlation can activate a globally optimal advantage, while the resulting advantage is nonmonotonic as a function of η. The exact null curve and the high-angle reentrant phase show that the resource can switch off again without changing the local Hamiltonians.

The result complements earlier work on adaptive channel discrimination [2-4], memory-limited discrimination [6], and multi-time correlation sensing [7]. The present contribution is not the general observation that adaptation or temporal memory can matter. It is an exact two-parameter resource geometry for a nonorthogonal Hamiltonian-recognition family, together with a proof that the sequential solution is already globally optimal over the GEN tester class.

The GEN closure also clarifies a nearby process-matrix result: adaptive and non-signalling strategies can coincide when the discriminated objects are causally ordered process matrices [8]. That theorem does not directly imply the present result because the roles of discriminated objects and tester affine spaces differ. Here the equality is obtained by an explicit task-specific GEN dual certificate reduction.

A natural next question is whether the polytope-to-elliptope geometry persists for more than two queries or for more general temporal-correlation kernels. Those extensions are outside the scope of the present work.

XIII. Methods and reproducibility

The exact theorems and constructive weak-correlation bounds are accompanied by independently executable verification scripts. The full-SEQ reduction, null-curve certificate, high-angle positivity proof, causal-comb normalization audit, GEN-dual symmetry reduction, and GEN-SEQ closure each have separate symbolic or numerical checks. Numerical optimization is used only for cross-checks and figure generation, not as a substitute for analytic proof.

The accompanying reproducibility archive contains the proof notes, verifier scripts, figure artifacts, numerical slice data, revision notes, and integrity metadata needed to reproduce the computational checks.

AI-assisted research disclosure. OpenAI ChatGPT (GPT-5.6 Sol) was used substantively during this research for symbolic-algebra workflow design, derivation checking, code generation and debugging, numerical cross-check design, literature organization, and manuscript drafting. The human author directed the research questions, reviewed the mathematical claims and generated artifacts, and assumes full responsibility for the accuracy, originality, and interpretation of the work. Analytic claims reported as exact theorems are supported by explicit derivations and independently executable verification scripts rather than by AI output alone. The weak-correlation statements

in Theorem 4 are intentionally stated as constructive lower bounds because no matching upper-bound derivation is included in the present archive.

Data Availability

All data needed to evaluate the conclusions are contained in the article, the Supplemental Material, and the accompanying reproducibility archive released with this Zenodo record. The numerical data underlying the optimized $\beta = 60^\circ$ slice are included as a machine-readable CSV file. No proprietary or restricted datasets are used.

Code Availability

All code used to verify symbolic identities and generate numerical cross-checks is included in the accompanying reproducibility archive released with this Zenodo record. The analytic theorems do not rely on numerical optimization for their validity.

Acknowledgments

OpenAI ChatGPT (GPT-5.6 Sol) also assisted with manuscript organization, English drafting, and language editing. The author reviewed and revised all AI-assisted text and is solely responsible for the final manuscript.

Author Contributions

Youfu Wang conceived the research problem, directed the analysis, reviewed and interpreted the mathematical results, curated the reproducibility materials, and prepared the manuscript with the disclosed AI assistance.

Competing Interests

The author declares no competing interests.

References

[1] C. Zhu, S. He, Y.-A. Chen, L. Zhang, and X. Wang, "Optimal Hamiltonian recognition of unknown quantum dynamics," *npj Quantum Information* **12**, 36 (2026), doi:10.1038/s41534-026-01182-6.

[2] A. W. Harrow, A. Hassidim, D. W. Leung, and J. Watrous, "Adaptive versus nonadaptive strategies for quantum channel discrimination," *Physical Review A* **81**, 032339 (2010), doi:10.1103/PhysRevA.81.032339.

[3] J. Bavaresco, M. Murao, and M. T. Quintino, "Strict Hierarchy between Parallel, Sequential, and Indefinite-Causal-Order Strategies for Channel Discrimination," *Physical Review Letters* **127**, 200504 (2021), doi:10.1103/PhysRevLett.127.200504.

[4] J. Bavaresco, M. Murao, and M. T. Quintino, "Unitary channel discrimination beyond group structures: Advantages of sequential and indefinite-causal-order strategies," *Journal of Mathematical Physics* **63**, 042203 (2022), doi:10.1063/5.0075919.

[5] G. Chiribella, G. M. D'Ariano, and P. Perinotti, "Memory Effects in Quantum Channel Discrimination," *Physical Review Letters* **101**, 180501 (2008), doi:10.1103/PhysRevLett.101.180501.

[6] T.-A. Ohst, S. Zhang, H. C. Nguyen, M. Plávala, and M. T. Quintino, "Characterising memory in quantum channel discrimination via constrained separability problems," *Quantum* **10**, 1988 (2026), doi:10.22331/q-2026-01-28-1988.

[7] S. Zhu, X. You, A. Romanenko, and A. Grassellino, "Two-Tooth Bosonic Quantum Comb for Temporal-Correlation Sensing," *Physical Review Letters* **137**, 100403 (2026), doi:10.1103/6738-xcgc.

[8] P. Lewandowska, Ł. Pawela, and Z. Puchała, "Strategies for single-shot discrimination of process matrices," *Scientific Reports* **13**, 3046 (2023), doi:10.1038/s41598-023-30191-0.